

Problem Set: Formal Languages & Automata Theory

Problem 1. Prove that the language of palindromes over an alphabet with at least two letters is not regular.

Problem 2. Prove that for all $a, b, k, r \in \mathbb{N}$, the following language over $\{0, 1, \$\}$ is regular:

$$L = \{\text{bin}(x) \$ \text{bin}(y) : (a \cdot x + b \cdot y) \equiv r \pmod{k}\}$$

Problem 3. Let L be a regular language. Prove that the following languages are also regular:

$$\frac{L}{2} = \{w : \exists u \text{ such that } |u| = |w| \text{ and } wu \in L\} \quad \sqrt{L} = \{w : ww \in L\}$$

Problem 4. Write context-free grammars for the following languages over $\Sigma = \{a, b\}$:

- (1) The set of words containing the same number of occurrences of a and b .
- (2) The set of words containing twice as many occurrences of a as occurrences of b .
- (3) The set of words of even length where the number of occurrences of b in even positions equals the number of occurrences of b in odd positions.

Problem 5. Give an algorithm to decide whether a given context-free grammar G generates an infinite language.

Problem 6. Let $\Sigma = \{a, b, \$\}$. Prove that the language $L = \{w\$w : w \in \{a, b\}^*\}$ is not context-free, but its complement is context-free.

Problem 7. A word is *square-free* if it contains no non-empty subword of the form vv (where $v \neq \epsilon$). Prove that a language L consisting solely of square-free words is context-free if and only if L is finite.

Problem 8. Is the language $L = \{w\$v^R : w, v \in \{a, b\}^*, v \text{ is a prefix and a suffix of } w\}$ context-free? Is its complement context-free?

Hints

- Hint 1.** Consider the set of prefixes $S = \{a^n b \mid n \geq 1\}$. Show that every pair of distinct strings $a^i b$ and $a^j b$ (with $i \neq j$) must belong to distinct equivalence classes under the Myhill-Nerode relation by appending a suffix that completes a palindrome for one but fails for the other.
- Hint 2.** Binary numbers can be processed sequentially either from left-to-right or right-to-left. Build a DFA whose states track the running value of $(a \cdot x + b \cdot y) \pmod k$. Process $\text{bin}(x)$ first to compute $a \cdot x \pmod k$, store this remainder in the finite state, and then update it as you read $\text{bin}(y)$ digit-by-digit.
- Hint 3.** For $\frac{L}{2}$, track two states simultaneously in a cross-product NFA: one state simulates w starting from q_0 , and the other state guesses an intermediate state q_m and simulates a word u of length $|u| = |w|$ ending in an accepting state $q_f \in F$. For $\sqrt{L}$, construct an NFA that guesses a triple of states (q_0, q_i, q_f) where q_0 is initial, $q_f \in F$, and q_i is the state reached after reading the first copy of w .
- Hint 4.** For (1), prove and use the following structural property: a word with equal a s and b s either starts and ends with opposing characters ($a \dots b$ or $b \dots a$) or can be split into two non-empty subwords that each independently have equal a s and b s. For (2), generalize (1) by defining a non-terminal that generates net $+2$ a s relative to b s. For (3), deconstruct words into length-2 blocks ($\Sigma^2 = \{aa, ab, ba, bb\}$) and notice that $ab \rightarrow +1$ net b in even positions, $ba \rightarrow -1$ net b in even positions, and $aa, bb \rightarrow 0$.
- Hint 5.** Convert G to Chomsky Normal Form (CNF) and analyze its variable dependency graph. After eliminating non-generating and unreachable symbols, construct a directed graph where vertices are variables and an edge $A \rightarrow B$ exists if $A \rightarrow BC$ or $A \rightarrow CB$. $L(G)$ is infinite if and only if there is a reachable variable A that can derive a terminal string and lies on a directed cycle ($A \xrightarrow{+} uAv$).
- Hint 6.** For L not context-free, use the Pumping Lemma for CFLs on $z = a^p b^p \$ a^p b^p$. For $\bar{L}$ context-free, express non-membership as a union of conditions: wrong format (does not contain exactly one $\$$), $|u| \neq |v|$ for $u \$ v$, or $|u| = |v|$ but u and v differ at some position i . Construct an NPDA (or CFG) that non-deterministically guesses which condition holds.
- Hint 7.** Assume L is infinite. By the CFL Pumping Lemma, there exists a sufficiently long word $z \in L$ that can be pumped as $uv^i wx^i y \in L$ for all $i \geq 0$. Examine the substring vwx , and show that pumping $i = 2$ (or higher) forces a repeated block vv to appear in $uv^2 wx^2 y$, contradicting the assumption that L contains only square-free words.
- Hint 8.** For L , rewrite the condition: v being a prefix and a suffix of w means $w = vuv$ for some $u \in \{a, b\}^*$, so $L = \{vuv \$ v^R : v, u \in \{a, b\}^*\}$. Intersect L with the regular language $R = a^* b \$ a^*$ and analyze the resulting language with the CFL Pumping Lemma. For $\bar{L}$, express non-membership in L as a union of simpler format and mismatch conditions.